
UNIT 14 QUALITATIVE INDEPENDENT VARIABLES IN OLS MODELS *

Structure

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14.0 OBJECTIVES

After going through this Unit you will be able to

- Apply Chow test for structural stability of a regression model;
- explain the nature of dummy variables;
- use dummy variables in regression models;
- test for structural stability in dummy variable models; and
- pool cross-sectional and time-series data by using dummy variables.

14.1 INTRODUCTION

In regression models considered so far, we assumed that there is no change in parameters within a sample. There could be situations, however, where there is a change (or shift) in a parameter within the sample. For example, assume that industrial output of India depends upon inputs such as capital and labour apart from technology and policy environment. After the economic liberalization in 1991, there is a major shift in the economic environment in India - there is an improvement in the level of technology and the level of competition. In short, we can say that there is a structural change in the Indian economy after economic liberalization. In such circumstances we cannot assume that the parameters

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remain the same for the pre-liberalisation period and the post-liberalisation period. A question before us is: How do we represent structural change in a regression model?

Let us consider another scenario. We assumed that the explanatory variables are numerical or quantitative in nature. This may not always be the case. There can be instances where the dependent variable (say, monthly earning) depends upon certain qualitative variable (say, gender). Can we include a qualitative variable such as gender as explanatory variable in a regression model? Yes, we can. A qualitative variable can be represented by a *dummy variable*, which assumes numerical values as we will see later in this Unit.

The dummy variables are also known as *binary variables*, *indicator variables*, *categorical variables*, and *dichotomous variables*. In this unit we consider the role of qualitative explanatory variables in regression analysis. We include dummy variables in regression models and show that it is a flexible tool for handling many interesting problems.

14.2 CHOW TEST FOR STRUCTURAL STABILITY

We begin with the Chow test for structural stability of a regression model. A structural change (also called structural break) occurs if the parameters underlying a relationship differ from one subset of the data to another. There may be several relevant subsets of data with the possibility of several structural breaks in a sample. For the moment we consider two subsets of n_1 and n_2 observations so that sample size is $n = n_1 + n_2$. Suppose, for example, we wish to investigate whether savings rate differs between rich households and poor households. Suppose we have n_1 poor households and n_2 rich households. An option before us is to estimate separate regression equations for the poor and the rich households. By comparing the estimated coefficients of both the regression models we can find out whether there is difference in savings rate of poor and rich households. Alternatively, we can estimate the regression model for the poor households and predict the savings rate for the rich households (or, *vice versa*). It is however not clear which choice should be made, and the two procedures might well yield different answers! Chow test gives concrete idea on the structural stability of a model.

Let us consider regression models for two groups as follows:

Rich households:

$$Y_t = \alpha_1 + \alpha_2 X_t + \varepsilon_{1t} \quad t = 1, 2, \dots, n_1 \quad \dots (14.1)$$

Poor households:

$$Y_t = \beta_1 + \beta_2 X_t + \varepsilon_{2t} \quad t = 1, 2, \dots, n_2 \quad \dots (14.2)$$

where Y is total savings of a household, X is personal income, the ε 's are the error terms in the two equations, n_1 and n_2 are the number of observations in the two sub-samples.

A structural change may mean that the two intercepts are different ($\alpha_1 \neq \beta_1$), or the two slopes are different ($\alpha_2 \neq \beta_2$), or both the intercept and the slopes are different, or any other suitable combination of the parameters. If there is no structural change (i.e., there is structural stability), we can combine all the n_1 and n_2 observations and estimate one savings function as

$$Y_t = \lambda_1 + \lambda_2 X_t + \varepsilon_t \quad \dots (14.3)$$

There are two basic assumptions underlying the Chow test.

- (a) $\varepsilon_{1t} \sim N(0, \sigma^2)$ and $\varepsilon_{2t} \sim N(0, \sigma^2)$, i.e., the two error terms are normally distributed with the same variance, and
- (b) ε_{1t} and ε_{2t} are independently distributed.

The Chow test proceeds as follows: We assume that regression equations for both the groups are different. This is our null hypothesis. Therefore, both the equations (14.1) and (14.2) need to be estimated separately. This is our unrestricted model. On the other hand, we restrict the above scenario and combine both the groups. We estimate equation (14.3) – this is our restricted model. Thus, the alternate hypothesis is our restricted model. The steps for carrying out the Chow test is given below.

- (i) Combine all the n_1 and n_2 observations and estimate regression model (14.3). Obtain restricted residual sum of squares, say RSS_R . The degree of freedom for this model is $(n_1 + n_2 - k)$, where k is the number of parameters in the model.
- (ii) Estimate regression models (14.1) and (14.2) individually. Obtain their residual sum of squares. Let us denote it by RSS_1 and RSS_2 respectively. The degrees of freedom for regression model (14.1) is $(n_1 - k)$ and that for regression model (14.2) is $(n_2 - k)$.
- (iii) Add these two residual sums of squares ($RSS_1 + RSS_2$) and obtain RSS_{UR} (unrestricted residual sum of squares). The degrees of freedom for RSS_{UR} is $(n_1 + n_2 - 2k)$.
- (iv) Apply the following formula to obtain the F-ratio:

$$F = \frac{(RSS_R - RSS_{UR})/k}{RSS_{UR}/(n_1 + n_2 - 2k)} \quad \dots (14.4)$$

which follows the F distribution with $DF = (k, n_1 + n_2 - 2k)$. If the F computed from (14.4) exceeds the critical value at the chosen level of confidence level, we reject the hypothesis that the regressions (14.1) and (14.2) are the same. In other words, we reject the hypothesis of structural stability.

Example 14.1

Let us consider the following hypothetical data on personal savings and personal income by poor and rich households (in Rupees thousand). We want to test whether the savings function is the same in both the groups.

Poor Households			Rich Households		
Sl. No.	Savings	Income	Sl. No.	Savings	Income
1	0.36	8.8	1	0.59	15.5
2	0.21	9.4	2	0.90	16.7
3	0.08	10.0	3	0.95	17.7
4	0.20	10.6	4	0.82	18.6
5	0.10	11.0	5	1.04	19.7
6	0.12	11.9	6	1.53	21.1
7	0.41	12.7	7	1.94	22.8
8	0.50	13.5	8	1.75	23.9
9	0.43	14.3	9	1.99	25.2

In the above data, we have $n_1 = n_2 = 9$.

For group 1, poor households

$$\hat{Y}_t = -0.2623 + 0.0470 X_t$$

(0.3054) (0.0266)

$$t = (-0.8719) \quad (1.7700)$$

$$R^2 = 0.3092, \quad \text{RSS}_1 = 0.1396, \quad DF = 7$$

For group 2, rich households:

$$\hat{Y}_t = -1.7502 + 0.1504 X_t$$

(0.3576) (0.0175)

$$t = (-4.8948) \quad (8.5749)$$

$$R^2 = 0.9131, \quad \text{RSS}_2 = 0.1931, \quad DF = 7$$

For combined data (restricted model):

$$\hat{Y}_t = -1.0821 + 0.1178 X_t$$

(0.1452) (0.0088)

$$t = (-7.4548) \quad (13.4316)$$

$$R^2 = 0.9185, \quad \text{RSS}_R = 0.5722, \quad DF = (9 + 9 - 2) = 16$$

Let us plug in the above data into the Chow test formula given at (14.4).

We have $RSS_{UR} = RSS_1 + RSS_2 = 0.1396 + 0.1931 = 0.3327$

$$F = \frac{(RSS_R - RSS_{UR})/k}{RSS_{UR}/(n_1 + n_2 - 2k)} = \frac{(0.5722 - 0.3327)/2}{0.3327/14} = \frac{0.11975}{0.0238} = 5.04$$

At the 5 percent level of significance, the critical value of $F_{2,14} = 3.74$. Since the observed F value of 5.04 exceeds this critical value, we reject the null hypothesis that the savings function in the two time periods is the same.

On the basis of the Chow test we can draw the conclusion that the savings functions in the two groups are different; but what is the reason for this difference? Is it due to the difference in the intercept values, or the slope values, or in both? The Chow test does not provide any clue on this. The dummy variable model discussed below helps us in finding an answer to this question.

14.3 THE NATURE OF DUMMY VARIABLES

In regression analysis the dependent variable is frequently influenced not only by variables that can be readily quantified on some well-defined scale (e.g., income, output, prices, costs, weights, etc.), but also by other variables that are essentially qualitative in nature (e.g., marital status, gender, religion, caste, etc.). For example, holding all other factors constant, IIT graduates are found to earn more than their counterparts from the regional engineering colleges in India. Similarly, studies in the US have reported that female college teachers earn less than their male counterparts. Whatever may be the reason for this disparity, qualitative variables like gender, institution of education, etc. do influence the dependent variable and should be included among the independent variables.

Since qualitative variables usually indicate the presence or absence of some attribute or quality, such as rural or urban, male or female, married or unmarried, etc. We can quantify such attributes by constructing artificial variables that take value 1 or 0, 1 indicating the presence (or possession) of a particular attribute and 0 the absence of it and vice versa. For example, 1 may indicate that the person is a male and 0 may indicate that the person is female; or 1 may indicate that a person is educated and 0 that he/she is not educated and so on. Such variables which assume values 0 and 1 are called Dummy Variables.

Like the quantitative variables, the dummy variables can be used in regression analysis very easily. There can be two cases: (i) a regression model may contain only dummy explanatory variables. This type of regression models is called *analysis of variance (ANOVA) models*. (ii) a regression model may contain both quantitative variables and dummy variables. This type of regression models is called the *analysis of variance (ANOVA) models*.

14.3.1 ANOVA Models

The following model is an example of ANOVA model

$$Y_i = \alpha_1 + \alpha_2 D_i + u_i \quad \dots (14.5)$$

where, Y_i = annual starting salary of a school-teacher

$$D_i = \begin{cases} 1 & \text{if male teacher} \\ 0 & \text{otherwise (i.e., female teacher)} \end{cases}$$

Model (14.5) is similar to an ordinary two variable regression model. The only difference is the use of a qualitative or dummy variable D instead of quantitative explanatory variable X . (From now on in the present unit we shall be using D to denote dummy variable). Assuming all the other factors such as age, years of experience, etc. to be constant, the model (14.5) may enable us to find out whether gender (i.e., being a male or female) makes any difference in a school teacher's salary. In other words, it would enable us to find out whether a male school teacher's salary is different from that of a female school teacher having the same qualifications and years of experience.

Inclusion of the disturbance term u_i in the model (14.5) satisfies the usual assumptions of the classical linear regression model (CLRM). We obtain from (14.5) that the mean salary of a female school-teacher is:

$$E(Y_i | D_i = 0) = \alpha_1 + \alpha_2(0) = \alpha_1 \quad \dots (14.6)$$

and the mean salary of a male school-teacher is:

$$E(Y_i | D_i = 1) = \alpha_1 + \alpha_2(1) = \alpha_1 + \alpha_2 \quad \dots (14.7)$$

In the above model the intercept term α_1 gives the mean annual salary of a female school-teacher. The slope coefficient α_2 tells us how much the mean salary of a male school-teacher differs from that of his female counterpart with $(\alpha_1 + \alpha_2)$ reflecting the mean annual salary of a male school-teacher.

We can also test for the following hypothesis: Is there discrimination in accordance to the gender of an individual while determining the salary of school-teachers. Here, we run by running OLS on the regression equation (14.5) and find out on the basis of t -test whether the estimated α_2 is statistically significant or not.

We will demonstrate the above with the help of a hypothetical example. Let us consider Table 14.1 which gives hypothetical data of annual salary of school-teachers.

Table 14.1: Annual Salary of School-teachers

Salary(Y) (Rs. 000)	Gender Dummy (D) (Male =1, Female = 0)
22.0	1
19.0	0
21.2	1
17.5	0
17.0	0
18.0	0
21.7	1
18.5	0
21.0	1
20.5	1

The OLS results corresponding to the regression model (14.5) are as follows:

$$Y_i = 18.00 + 3.28 D_i \quad \dots (14.8)$$

$$\text{Standard error} \quad (0.32) \quad (0.44)$$

$$\text{t-statistic} \quad (57.74) \quad (7.44)$$

$$R^2 = 0.8737$$

From (14.8) we see that the estimated mean salary of female school-teachers ($\hat{\alpha}_1$) is Rs.18,000 and that of male school-teachers ($\hat{\alpha}_1 + \hat{\alpha}_2$) is Rs.21,280.

From the calculated t-statistic in regression model (14.8), it is easy to verify that $\hat{\alpha}_2$ is statistically significant (i.e., significantly different from zero), suggesting a difference in starting salaries of male and female school-teachers. If we represent the regression model (14.8) a diagram, we obtain a step-function of the type given in Fig. 14.1.

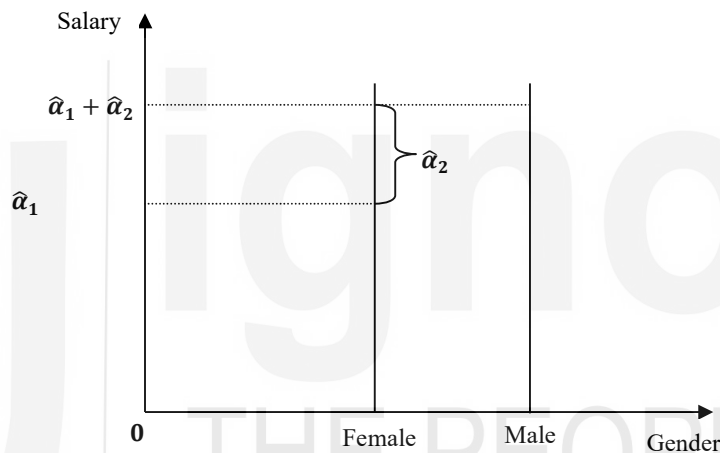


Fig. 14.1: Salary of Male and Female Teachers (Rs. '000)

ANOVA models such as those in (14.8), although frequently used in fields such as sociology, education, market research, etc. are not common in economics. Regression models in most economic research consist of explanatory variables that are both quantitative and qualitative in nature. Such type of regression models which contain both qualitative and quantitative explanatory variables are known as *analysis of covariance (ANCOVA) models*.

14.3.2 ANCOVA Models

As mentioned earlier, a regression model can have both qualitative (dummy) variables and quantitative variables. Such models are called ANCOVA models. Let us now consider the following as an example of the ANCOVA model:

$$Y_i = \alpha + \beta D_i + \gamma X_i + u_i \quad \dots (14.9)$$

where, Y_i = annual salary of a school-teacher

X_i = years of teaching experience

D_i = 1 if male teacher

0 otherwise (i.e., female teacher)

Regression model (14.9) is similar to (14.5) except that it contains a quantitative variable (years of teaching, X_i) in addition to the qualitative variable (gender of the teacher, D_i). The qualitative variable has two categories, namely, male and female. Thus, the model has two explanatory variables: one qualitative and one quantitative.

What does model (14.9) mean/imply? Assuming, as usual, $E(u_i) = 0$, we see that the mean salary of a female school-teacher is

$$E(Y_i | D_i = 0) = \alpha + \gamma X_i \quad \dots (14.10)$$

and the mean salary of male school-teacher is

$$E(Y_i | D_i = 1) = (\alpha + \beta) + \gamma X_i \quad \dots (14.11)$$

From the model (14.9) we see that the male and female school-teachers' salary function in relation to the years of experience have the same slope (γ) but different intercepts. In other words, in the model, it is assumed that the level of mean salary of male school-teacher is different from that of his female counterpart but the rate of change in the mean salary by years of experience (represented by the slope term γ) is the same for both male and female school-teachers. Geometrically we can represent models (14.6) and (14.7) as shown in Fig. 14.2 (in the figure α is assumed to be positive, i.e., $\alpha > 0$).

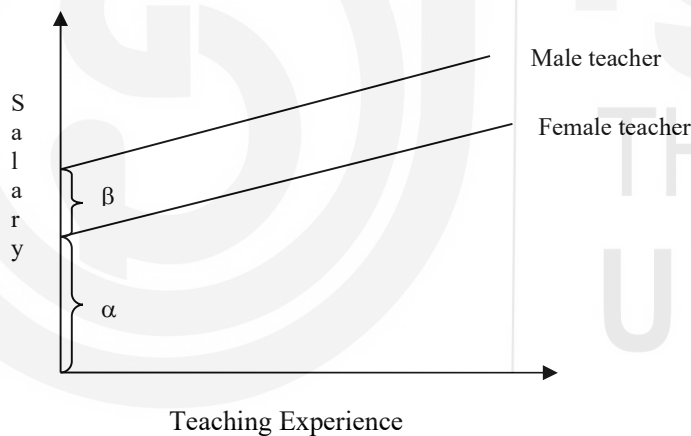


Fig. 14.2: Annual Salary of Male and Female Teachers

Just as in regression (14.8), here also we can use the t -test to test for the hypothesis that male and female school-teachers have the same mean annual salary.

Before proceeding further, it is essential to discuss some of the important features of the dummy variables:

- (i) In dummy variable models discussed so far, we have introduced only one dummy variable, D_i to distinguish between two categories. For example, male and female with $D_i = 1$ denoting male and $D_i = 0$ denoting female. Now what happens if instead of one dummy variable two dummy variables

D_{1i} and D_{2i} are introduced in the model, one each for male and female? Model (14.9) can now be written as

$$Y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \gamma X_i + u_i \quad \dots (14.12)$$

where, Y_i and X_i are as defined before, and

$$D_{1i} = 1 \quad \text{if a male teacher}$$

$$0 \quad \text{otherwise}$$

$$D_{2i} = 1 \quad \text{if a female teacher}$$

$$0 \quad \text{otherwise}$$

Due to perfect collinearity between D_1 and D_2 (i.e., perfect linear relationship) the regression model (14.12) cannot be estimated. This can be explained with the help of the data given in Table 14.2.

Table 14.2: Annual Salary of School Teachers

		Intercept	D_1	D_2	X
Male	Y_1	1	1	0	X_1
Male	Y_2	1	1	0	X_2
Female	Y_3	1	0	1	X_3
Male	Y_4	1	1	0	X_4
Female	Y_5	1	0	1	X_5

From Table 14.2 it is easy to verify that D_1 and D_2 are perfectly collinear as $D_1 = (1 - D_2)$ or $D_2 = (1 - D_1)$. We know from Unit 10 that in the case of perfect multicollinearity, it is not possible to estimate the various parameters.

There are, however, several ways of resolving this problem but the simplest one is by using dummy variables as we had done in model (14.5). We should use only one dummy variable if there are two categories of a qualitative variable.

Rule of Thumb: If a qualitative variable has 'm' categories, introduce '(m - 1)' dummy variables.

Thus, if a qualitative variable has 4 characteristics, we introduce only 3 dummy variables. If this rule is not followed, we will fall into what is known as the *dummy variable trap*, i.e., a situation of perfect multicollinearity.

- (ii) The assignment of values 0 and 1 to two categories such as rural and urban, or educated and uneducated, etc., is arbitrary. For example, in our model (14.5) instead of assigning 1 to male teacher and 0 to female teacher we could have assigned the value 1 to female teacher and 0 to male teacher

(and the coefficients would change accordingly). In such a case, how do we interpret the results? While interpreting the results of the models that use dummy variables, it is critical to know how the values 1 and 0 are assigned. The category that is assigned value 0 is referred to as the ‘base category’ or ‘benchmark category’ and all the comparisons are made with reference to this category. In model (14.5) female school-teacher, which is assigned value 0, is the base or benchmark category.

- (iii) The coefficient attached to the dummy variable (for example β in model 14.5) is referred to as the differential intercept coefficient because it tells by how much the value of the intercept term of the category that receives value 1 differs from that of the base category.

Check Your Progress 1

- 1) Under what circumstances is the use of dummy variable suggested? Give some practical examples where dummy variables can be fitted.

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- 2) Suppose an explanatory variable is divided into 4 categories and you plan to fit 4 dummies in the regression equation. Is it possible to estimate the equation?

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14.4 USE OF MORE THAN ONE DUMMY VARIABLE

We can extend the analysis in the previous section to handle more than one qualitative variable with ease. Consider once again the example of salary of school-teacher which was used in the previous section. In this section we assume that the salary of a school-teacher to be dependent not only on the years of experience and gender of the teacher but also on the type of education of the teacher. In the present example it is assumed that the type of education has two categories, convent educated and non-convent educated. Incorporating the variable type of education of the teacher in regression model (14.5) we can write it as:

$$Y_i = \alpha + \beta D_{1i} + \gamma D_{2i} + \delta X_i + u_i \quad \dots (14.13)$$

where, Y_i and X_i are as defined before and